

Rh FACTOR

- **Definition** : Rh factor is an antigen present on RBCs.
- **Discovery** : Discovered by **Landsteiner and Wiener** (*first discovered in Rhesus monkey*).
- **Antigenicity** : Only **D-antigen** is strongly antigenic in humans.
 - a. **Rh +ve** : 85% of people (D-antigen present)
 - b. **Rh -ve** : 15% of people (D-antigen absent)

Difference from ABO Group System :

Antigen D does **not** have a naturally occurring antibody (*anti-D*).

If Rh +ve blood is transfused into an Rh -ve person, **anti-D is developed in that person**.

INHERITANCE OF Rh ANTIGEN

- Rhesus factor is an **inherited dominant factor**.
- **Genotypes** :—
 - a. **Homozygous Rhesus +ve** : DD
 - b. **Heterozygous Rhesus +ve** : Dd
 - c. **Homozygous Rhesus -ve** : dd

Father Genotype	Mother Genotype	Offspring Outcome
Father DD (Rh⁺) (Homozygous)	Mother dd (Rh⁻)	100% offspring are Dd (Rh⁺ - Heterozygous)
Father Dd (Rh⁺) (Heterozygous)	Mother dd (Rh⁻)	50% Dd (Rh⁺) and 50% dd (Rh⁻)
Father dd (Rh⁻) (Homozygous)	Mother dd (Rh⁻)	100% offspring are dd (Rh⁻)

TRANSFUSION REACTIONS DUE TO Rh

INCOMPATIBILITY

1st Exposure (Rh+ blood to Rh- person)

Rh- person receives Rh+
blood (1st time)

Rh antibodies (anti-D)
develop slowly within 1
month

Transfused RBCs present in
recipient's blood
agglutinated

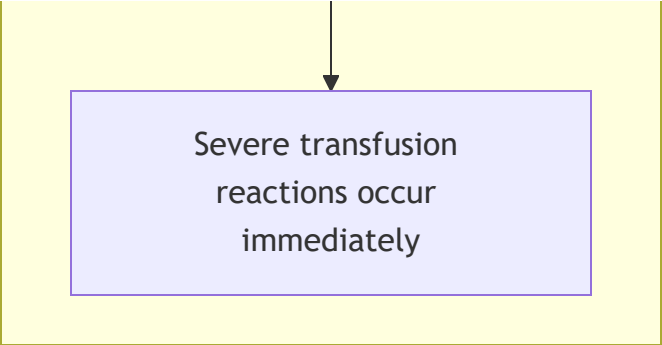
Agglutinated cells lysed by
macrophages

Antibodies developed in
recipient remain forever

2nd Exposure (Subsequent Transfusion)

Person receives Rh+ blood
(2nd time)

Immediate agglutination of
donor RBCs



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Severe transfusion
reactions occur
immediately